Math 281 Final Concept Sheet — Master Workflows

Master Router (Fast Recognition)

- Sphere / quadric / plane / line / distance? Use geometry formulas.
- Vector curve? Differentiate componentwise. Speed = $\|r'\|$, length = $\int \|r'\|\,dt$.
- Limit in 2 vars? Test two paths. Same lines does not prove existence.
- First/Second partials? Differentiate carefully. Keep mixed terms straight.
- Chain rule / du / related rates? Write base formula first, then differentiate.
- Intersection curve tangent? Use $\nabla F \times \nabla G$ for tangent direction.
- Tangent plane / normal line? Separate $z = f(x,y)$ from $F(x,y,z) = c$.
- Directional derivative / max rate? ∇f, normalize direction, then dot product.
- Local extrema? Solve $f_x = f_y = 0$, then second derivative test.
- Absolute extrema on closed region? Interior crit pts + every boundary + endpoints.
- Constraint given? Use Lagrange multipliers: $\nabla f = \lambda \nabla g$.

Exam Strategy: Label the type first. Do not start computing blindly. Use the shortest valid route. If a line-path already gives parameter-dependent answer, stop. For optimization, write the full pipeline every time. For geometry, normals and direction vectors are everything.

Geometry, Lines, Planes (Ch.12)

Vector basics: $\|u\| = \sqrt{u_1^2 + u_2^2 + u_3^2}$, $\hat{u} = u/\|u\|$.

Dot: $u \cdot v = u_1v_1 + u_2v_2 + u_3v_3$,

$\cos\theta = \dfrac{u \cdot v}{\|u\|\,\|v\|}$.

- Orthogonal: $u \cdot v = 0$ Parallel: $u = \lambda v$
- Scalar comp of b along a: $\text{comp}_a b = \dfrac{b \cdot a}{\|a\|}$
- Proj of b onto a: $\text{proj}_a b = \dfrac{b \cdot a}{\|a\|^2} a$

Cross: $\|u \times v\| = \|u\|\,\|v\| \sin\theta$

- parallelogram area = $\|u \times v\|$
- triangle area = $\frac{1}{2}\|u \times v\|$
- normal to plane from 3 points: $(B - A) \times (C - A)$

Triple product: $V = |a \cdot (b \times c)|$ gives parallelepiped volume.

Lines: $r(t) = r_0 + td$

Parametric: $x = x_0 + at$, $y = y_0 + bt$, $z = z_0 + ct$

Symmetric: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ (if comp is 0, keep constant)

Planes: Vector: $n \cdot (\langle x,y,z\rangle - \langle x_0,y_0,z_0\rangle) = 0$

Scalar: $Ax + By + Cz = D$ with normal $n = \langle A,B,C\rangle$.

Distance Formulas:

- Point to plane: $d = \dfrac{|Ax_0+By_0+Cz_0-D|}{\sqrt{A^2+B^2+C^2}}$
- Point to line thru A, B: $d = \dfrac{\|(P-A)\times(B-A)\|}{\|B-A\|}$
- Parallel planes D_1, D_2: $d = \dfrac{|D_2-D_1|}{\sqrt{A^2+B^2+C^2}}$
- Skew lines ($L_1 : a + tu$, $L_2 : b + sv$): $d = \dfrac{|(b-a)(u\times v)|}{\|u\times v\|}$. If $u \times v = 0$, use point-to-line.

Ch. 12 Checklists:

- Sphere from center + point: $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$, r = distance to point.
- Plane thru 3 points: form two dir vectors → cross for normal → plug point into point-normal form.
- Plane thru point containing line: line gives one dir vector → point on line to given point gives second → cross for normal.
- Line-plane intersection: plug line parametrics into plane → solve for parameter → substitute back.
- Parallel / skew lines: compare dir vectors → if not parallel, solve simultaneously. No common point = skew.
- Angle between planes: $\cos\theta = \frac{n_1 n_2}{\|n_1\|\|n_2\|}$. Use

acute.

Surface ID / Quadratic Surfaces:

- Missing variable ⇒ cylinder in that direction
- sphere: $(x-a)^2+(y-b)^2+(z-c)^2=r^2$
- ellipsoid: $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$
- elliptic paraboloid: $z=ax^2+by^2+k$
- hyperbolic paraboloid: $z=ax^2-by^2+k$
- 1-sheet hyperboloid: $+,+,-=1$
- 2-sheet hyperboloid: $+,-,-=1$
- cone: $z^2=ax^2+by^2$

Fast ID tool: set $x=c$, $y=c$, $z=c$ and inspect traces.

Vector Curves (Ch.13)

Space curve: $r(t)=\langle x(t),y(t),z(t)\rangle$.
Velocity: $v=r'(t)$ **Accel:** $a=r''(t)$ **Speed:** $\|r'(t)\|$
Rules: $(cu)'=cu'$, $(u\cdot v)'=u'\cdot v+u\cdot v'$, $(u\times v)'=u'\times v+u\times v'$

Ch. 13 Checklists:

- **Domain/Limit:** intersect domains / componentwise limits.
- **Curve length:** compute $r'(t)$ → compute $\|r'(t)\|$ → integrate $\int_a^b \|r'(t)\|\,dt$.
- **Reparametrize:** compute speed → integrate to get $s(t)$ → invert to get $t(s)$ → substitute into $r(t)$.
- **Curve ID:** Line? All components affine (linear). Parabola? One quadratic, another linear. Intersection? Parameterize easy piece first.

Tangent, Normal, Binormal, Curvature:

$$T=\frac{r'}{\|r'\|},\qquad N=\frac{T'}{\|T'\|},\qquad B=T\times N$$

$$\kappa=\frac{\|T'\|}{\|r'\|}=\frac{\|r'\times r''\|}{\|r'\|^3}$$

Curvature Checklists:

- Curve $y=f(x)$: $\kappa=\frac{|f''(x)|}{(1+[f'(x)]^2)^{3/2}}$
- Parametric: $\kappa=\frac{|x'y''-y'x''|}{(x'^2+y'^2)^{3/2}}$
- Osculating plane: normal B. Checklist: compute r', r'' → find $B \parallel r'\times r''$ → use point-normal plane eq.
- Normal plane: normal vector T
- Osculating circle: same tangent, normal, curvature, radius $=1/\kappa$

Acceleration Decomposition:

$a_T=\frac{v\cdot a}{\|v\|}$, $\quad a_N=\sqrt{\|a\|^2-a_T^2}$.
Meaning: a_T changes speed, a_N bends path.

Limits, Partials, Rates (Ch.14)

Limits in 2 vars: To show DNE, pick two paths with diff answers.

- start with lines: $y=mx$
- then try curved paths: $y=x^2$, $y=kx^2$, $x=y^2$
- polar helps when expression has x^2+y^2

Patterns: If denom has xy, test $y=x$ and $y=-x$. If denom has x^2+y^2, compare num order. If polar gives leftover θ, DNE.
Example: $\frac{2xy}{x^2+2y^2}$ → try $y=mx$ → $\frac{2x(mx)}{x^2+2m^2x^2}=\frac{2m}{1+2m^2}$. Depends on m, so DNE (Limit cue: if result depends on m, limit DNE immediately).
Continuity & Linearization: Continuous if limit exists and equals function value.
Differentiable means locally approximated by a linear map:
$f(a+\Delta x,b+\Delta y)\approx f(a,b)+f_x\Delta x+f_y\Delta y$.
Linearization: $L(x,y)=$ $f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$
Second Partials + Clairaut: keep first partials easy to differentiate. Clairaut: if continuous nearby, $f_{xy}=f_{yx}$.
Example: $f(x,y)=4x^3-xy^2 \Rightarrow f_x=12x^2-y^2$, $f_y=-2xy \Rightarrow f_{xx}=24x$, $f_{xy}=-2y$, $f_{yx}=-2y$, $f_{yy}=-2x$.
Differentials: $du=u_x\,dx+u_y\,dy$
Example:
$u=\ln(1+se^{2t}) \Rightarrow du=\frac{1}{1+se^{2t}}\left(e^{2t}\,ds+2se^{2t}\,dt\right)$
Chain Rule Checklist: compute f_x, f_y, f_z → compute x', y', z' → substitute all expressions.
$\frac{du}{dp}=f_x\frac{dx}{dp}+f_y\frac{dy}{dp}+f_z\frac{dz}{dp}$.
Implicit Differentiation: If $F(x,y)=0$,

$\frac{dy}{dx} = -\frac{F_x}{F_y}$. For surface $F(x,y,z) = 0$, $z_x = -\frac{F_x}{F_z}$, $z_y = -\frac{F_y}{F_z}$.

Related rates: write physical formula first, then differentiate in time. Do not plug numbers until after differentiating.

Example: $A = \frac{1}{2}xy\sin\theta \Rightarrow \frac{dA}{dt} = \frac{1}{2}\left(y\sin\theta\frac{dx}{dt} + x\sin\theta\frac{dy}{dt} + xy\cos\theta\frac{d\theta}{dt}\right)$.

Gradients, Tangents, Normals

Gradient & Directional Derivative: $D_{\hat{u}}f(P) = \nabla f(P)\cdot\hat{u}$.

Checklist: compute gradient → eval at point → make direction unit length → dot product.

Toward point Q from P: use $Q - P$. Example: from $(-2,0)$ toward $(2,-3)$ use direction $\langle 4,-3\rangle$, then divide by 5.

Max Rate: $\max D_{\hat{u}}f = \|\nabla f\|$. Direction of max inc = $\nabla f/\|\nabla f\|$. Max dec = $-\nabla f/\|\nabla f\|$.

Example: Max inc of $f(x,y,z) = ze^{xy}$ at $(0,1,2)$. $\nabla f = \langle zye^{xy}, zxe^{xy}, e^{xy}\rangle$. Evaluate first. Exam trap: direction = vector, rate = magnitude.

Tangent vs Normal Quick Split:

- Tangent plane to surface: plane that touches surface locally
- Normal line to surface: line perpendicular to tangent plane
- Tangent line to curve: line in curve direction
- Normal plane to curve: plane perpendicular to tangent line

For graph $z = f(x,y)$ at (a,b,z_0):

- Tangent plane: $z - z_0 = f_x(a,b)(x-a) + f_y(a,b)(y-b)$
- Normal vector: $\langle f_x(a,b), f_y(a,b), -1\rangle$
- Normal line: $(x,y,z) = (a,b,z_0) + t\langle f_x(a,b), f_y(a,b), -1\rangle$

For level surface $F(x,y,z) = c$ at P:

- Tangent plane: $\nabla F(P)\cdot\langle x-x_0, y-y_0, z-z_0\rangle = 0$
- Normal vector: $\nabla F(P)$
- Normal line: $(x,y,z) = P + t\,\nabla F(P)$

Review cue: $z = 3x^2 - y^2 + 2x$ and $z = e^x\cos y$ are graph form. $x^2 + 2y^2 - 3z^2 = 3$, $xy + yz + zx = 3$, and $\sin(xyz) = x + 2y + 3z$ are level surfaces.

Parallel tangent plane prob: If tang plane to $F = c$ is parallel to $Ax + By + Cz = D$, normals are parallel: $\nabla F(P) = \lambda\langle A,B,C\rangle$. Solve with surface equation.

Example: $x^2 + 4y^2 - z^2 = 4$, plane $2x + 2y + z = 5$. Set $\nabla F = \langle 2x, 8y, -2z\rangle = \lambda\langle 2,2,1\rangle$, plug into surface.

Intersection tangent line: If curve is intersection of $F = 0$ and $G = 0$, then tangent direction is

$$d = \nabla F(P) \times \nabla G(P)$$

So tangent line is

$$(x,y,z) = P + t\,d$$

Example: surface $z = 2x^2 - y^2$ and plane $z = 4$. Rewrite as $F = z - 2x^2 + y^2$, $G = z - 4$. Cross product at point.

Optimization

Local Extrema Checklist: solve $f_x = f_y = 0$ → compute f_{xx}, f_{yy}, f_{xy} → classify with $D = f_{xx}f_{yy} - (f_{xy})^2$.

- $D > 0, f_{xx} > 0$: min $D > 0, f_{xx} < 0$: max
- $D < 0$: saddle $D = 0$: inconclusive

Exam trap: solve system completely, don't stop at one point.

Absolute Extrema Checklist: interior crit points → reduce each boundary to one var → compare all values.

Example Triangle $(0,0)$, $(0,6)$, $(6,0)$: edge 1 ($x = 0$), edge 2 ($y = 0$), edge 3 ($x + y = 6$). Sub into f, solve 1-var extrema.

Example Disk $x^2 + y^2 \le 4$: interior ($f_x = f_y = 0$), boundary (use $x = 2\cos t,\ y = 2\sin t$ or $x^2 + y^2 = 4$).

Closed / Bounded / EVT:

- Closed set: contains boundary points
- Bounded set: fits inside some large disk/ball
- EVT: continuous on closed and bounded set ⇒ absolute max/min exist

Lagrange Multiplier Method:

$$\nabla f = \lambda \nabla g, \qquad g = c$$

2-variable form:

$$f_x = \lambda g_x, \qquad f_y = \lambda g_y, \qquad g(x,y) = c$$

3-variable form:

$$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad f_z = \lambda g_z, \quad g(x,y,z) = c$$

Solve all together, then compare values of f at all candidate points.

Fact / Traps / T/F Bank

Algebra & Exam Traps:

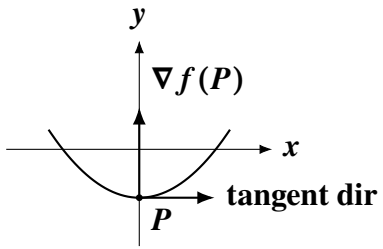
- **normalize direction vectors**
- **do not lose minus signs in f_{xy}**
- **for level surfaces, point must satisfy surface first**
- **for tangent plane parallelism, compare normals**
- **for closed regions, boundaries matter**
- **when plugging line into plane, solve parameter carefully**

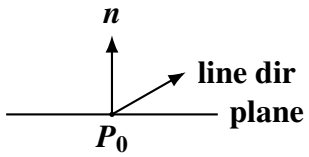
Tiny Vocab Facts:

- **Normal vector is perp to a plane or tangent plane.**
- **Tangent vector points along a curve: $r'(t)$.**
- **Normal vector to level surface $F = c$ is ∇F.**
- **Normal plane to a curve has normal T.**
- **Osculating plane to a curve has normal B.**
- **Critical point: where $\nabla f = 0$ or partials DNE.**

False unless special case:
$|u + v| = |u| + |v|$, $|u \cdot v| = |u|\,|v|$,
$|u \times v| = |u|\,|v|$, $u \times v = v \times u$,
$u \times (v \times w) = (u \times v) \times w$, **if f_x, f_y exist then f diff, if all line-path limits agree then full limit exists.**

True core facts:
$u \cdot (v \times w) = (u \times v) \cdot w$, $(u \times v) \cdot u = 0$,
Cauchy-Schwarz: $|u \cdot v| \le \|u\|\,\|v\|$, **if** $\|r(t)\| = 1$, **then** $r \cdot r' = 0$, **if $\kappa(t) = 0$ for all t, curve is a line.**





Parameterization Quick Guide
Line through point P with direction d: $(x,y,z) = P + t\,d$. **Example: through $(1,2,3)$ dir $\langle 2,-1,4\rangle$:** $x = 1 + 2t$, $y = 2 - t$, $z = 3 + 4t$.
Circle / cylinder $(x^2 + y^2 = r^2)$: Use $x = r\cos t$, $y = r\sin t$.
Ellipse $(x^2/a^2 + y^2/b^2 = 1)$: Use $x = a\cos t$, $y = b\sin t$.
Intersection curve recipe: 1. Parameterize the easier equation (often a circle). 2. Let t be the parameter. 3. Substitute into the other equation.
Example: $x^2 + y^2 = 16$, $x + z = 5$. Use $x = 4\cos t$, $y = 4\sin t$. **Then** $z = 5 - 4\cos t$.
Tangent line to intersection of surfaces $F = 0$, $G = 0$: Direction = $\nabla F \times \nabla G$. Line: $(x,y,z) = P + t(\nabla F \times \nabla G)$.